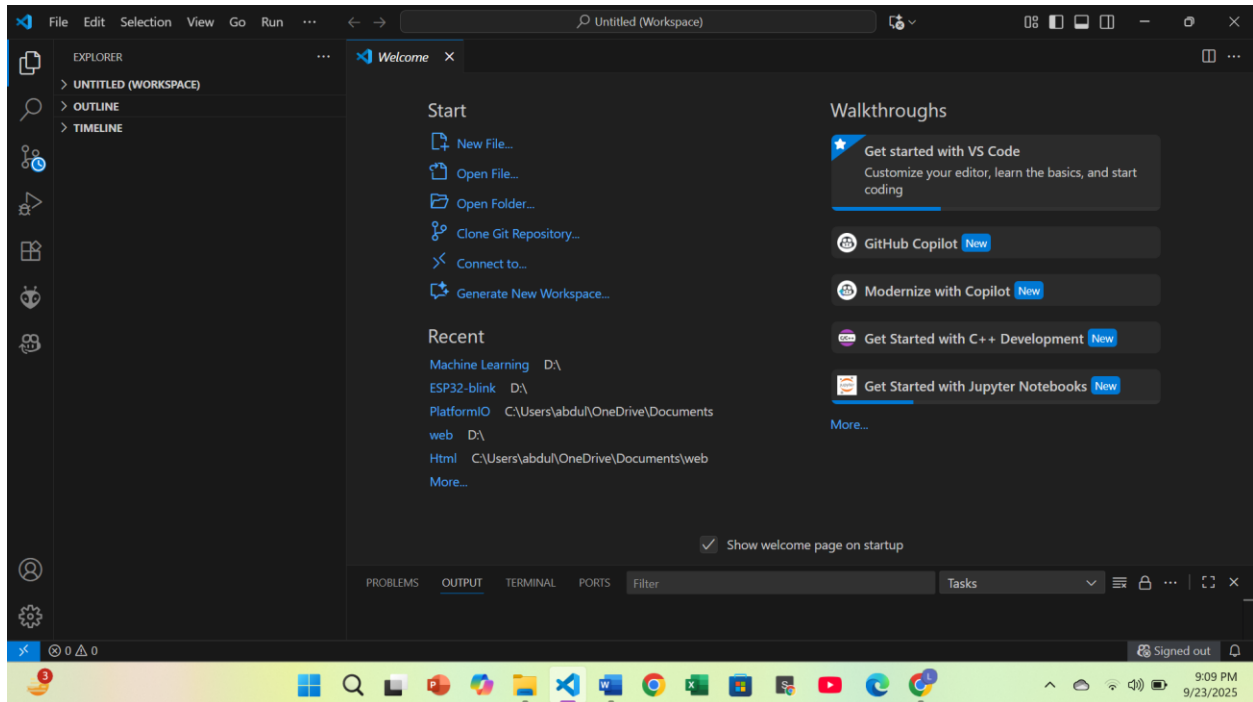
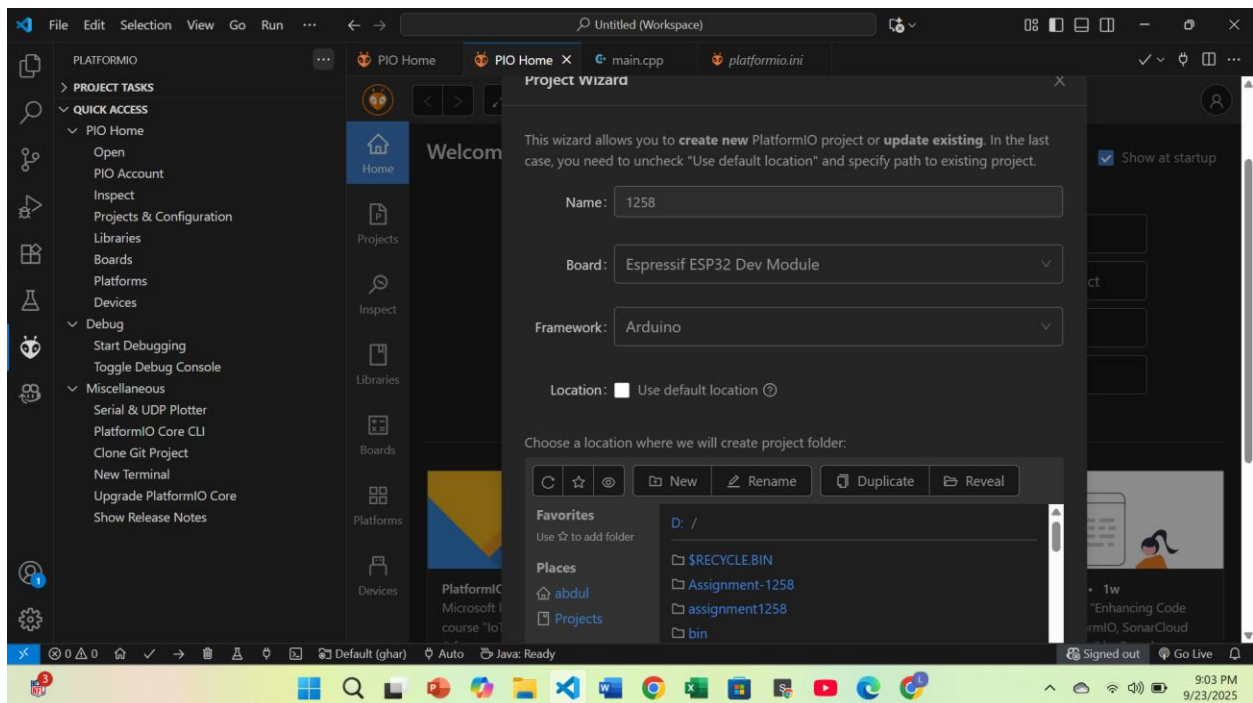


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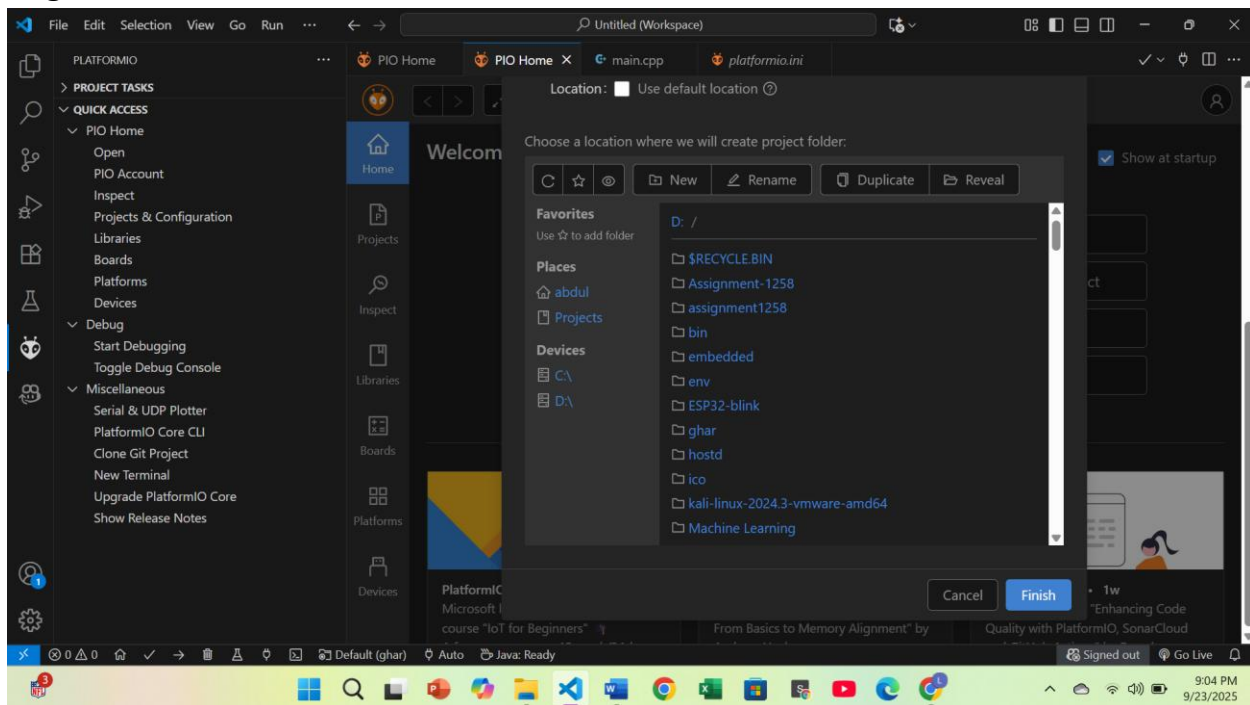
1) Open VScode



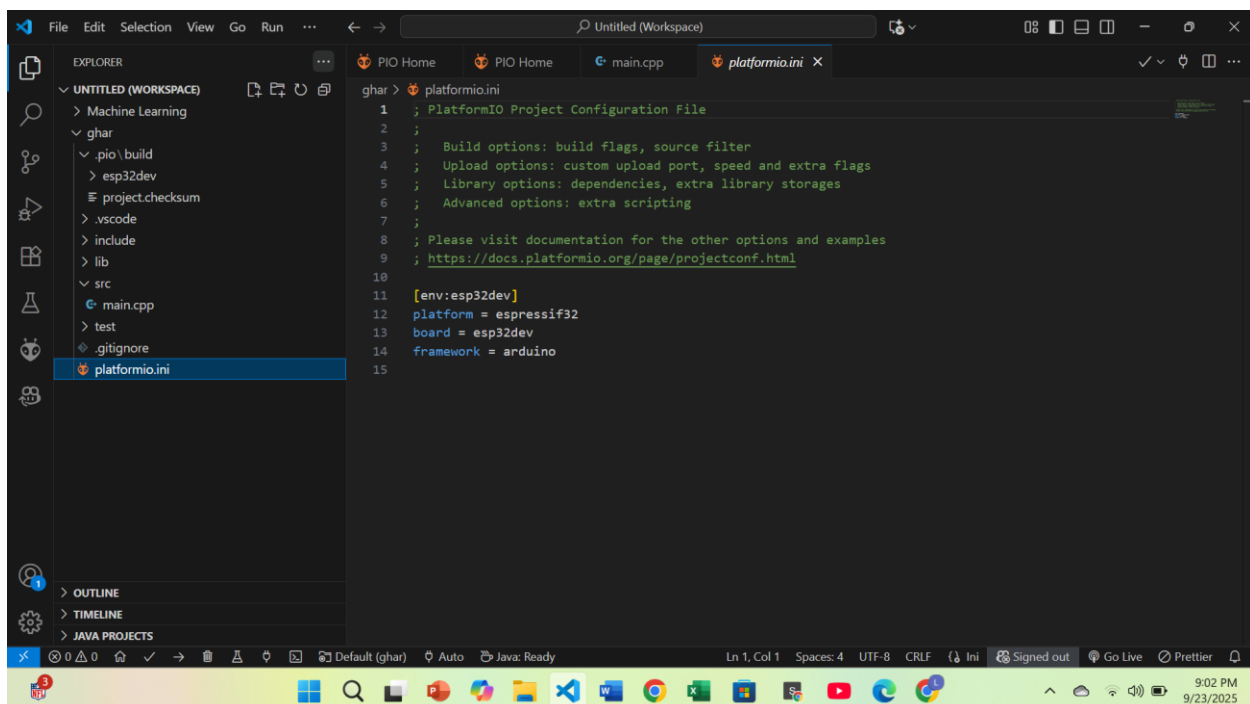
2) Create Project



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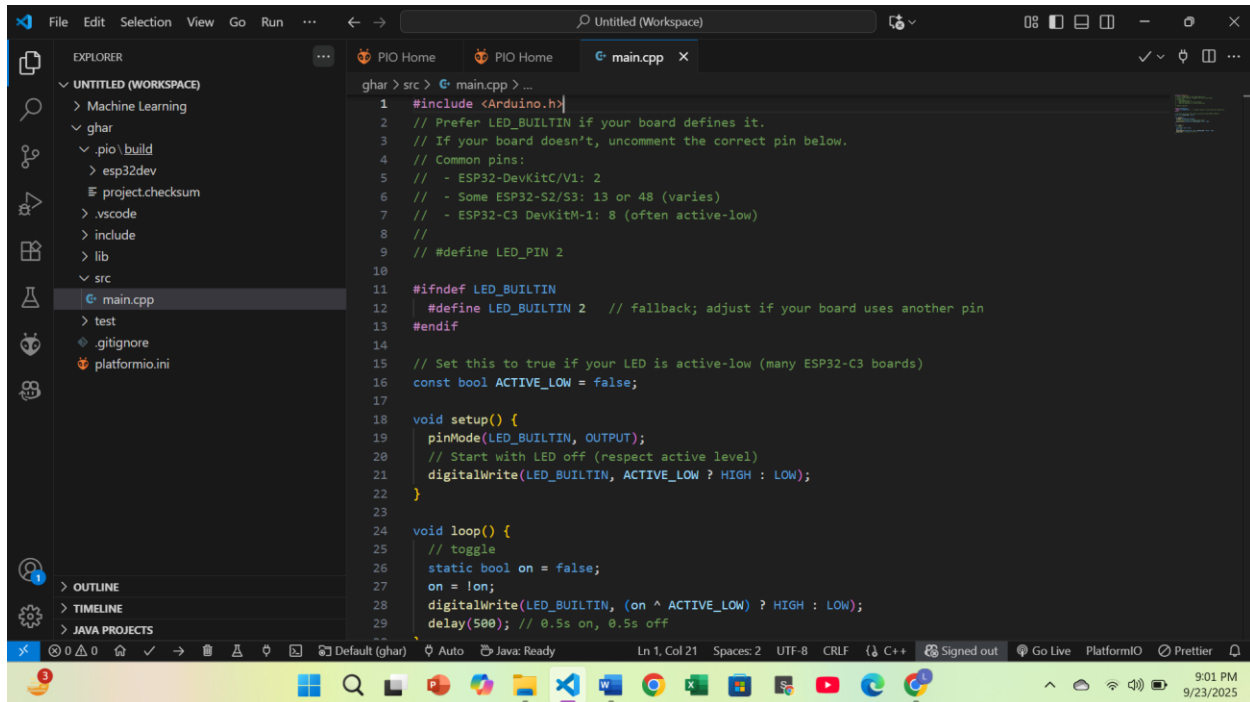


3) Platformio.ini



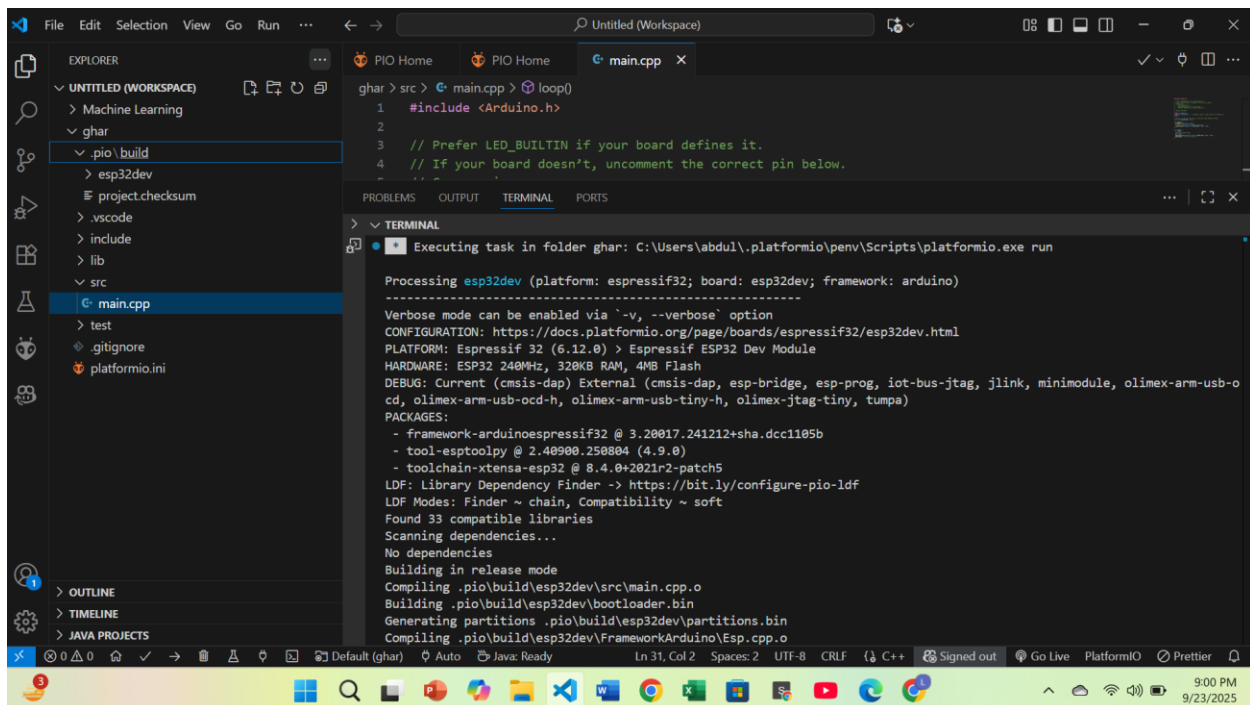
Name: Laiba Taseen
Reg No: 23-NTU-CS-1258

4) Blink Code



```
1 #include <Arduino.h>
2 // Prefer LED_BUILTIN if your board defines it.
3 // If your board doesn't, uncomment the correct pin below.
4 // Common pins:
5 // - ESP32-DevKitC/V1: 2
6 // - Some ESP32-S2/S3: 13 or 48 (varies)
7 // - ESP32-C3 DevKitM-1: 8 (often active-low)
8 //
9 // #define LED_PIN 2
10
11 #ifndef LED_BUILTIN
12 #define LED_BUILTIN 2 // fallback; adjust if your board uses another pin
13 #endif
14
15 // Set this to true if your LED is active-low (many ESP32-C3 boards)
16 const bool ACTIVE_LOW = false;
17
18 void setup() {
19   pinMode(LED_BUILTIN, OUTPUT);
20   // Start with LED off (respect active level)
21   digitalWrite(LED_BUILTIN, ACTIVE_LOW ? HIGH : LOW);
22 }
23
24 void loop() {
25   // toggle
26   static bool on = false;
27   on = !on;
28   digitalWrite(LED_BUILTIN, (on ^ ACTIVE_LOW) ? HIGH : LOW);
29   delay(500); // 0.5s on, 0.5s off
```

5) Build, upload and monitor



```
ghar > src > main.cpp > loop()
1 #include <Arduino.h>
2
3 // Prefer LED_BUILTIN if your board defines it.
4 // If your board doesn't, uncomment the correct pin below.
```

PROBLEMS OUTPUT TERMINAL PORTS

Executing task in folder ghar: C:\Users\abdu\platformio\penv\Scripts\platformio.exe run

Processing esp32dev (platform: espressif32; board: esp32dev; framework: arduino)

Verbose mode can be enabled via '-v, --verbose' option

CONFIGURATION: <https://docs.platformio.org/page/boards/espressif32/esp32dev.html>

PLATFORM: Espressif 32 (6.12.0) > Espressif ESP32 Dev Module

HARDWARE: ESP32 240MHz, 320KB RAM, 4MB Flash

DEBUG: Current (cmsis-dap) External (cmsis-dap, esp-bridge, esp-prog, iot-bus-jtag, jlink, minimodule, olimex-arm-usb-ocd, olimex-arm-usb-ocd-h, olimex-arm-usb-tiny-h, olimex-jtag-tiny, tumpa)

PACKAGES:

- framework-arduinoespressif32 @ 3.20017.241212+sha.dcc1105b
- tool-esptoolpy @ 2.40900.250804 (4.9.0)
- toolchain-xtensa-esp32 @ 8.4.0+202112-patch5

LDF: Library Dependency Finder -> <https://bit.ly/configure-pio-ldf>

LDF Modes: Finder ~ chain, Compatibility ~ soft

Found 33 compatible libraries

Scanning dependencies...

No dependencies

Building in release mode

Compiling .pio\build\esp32dev\src\main.cpp.o

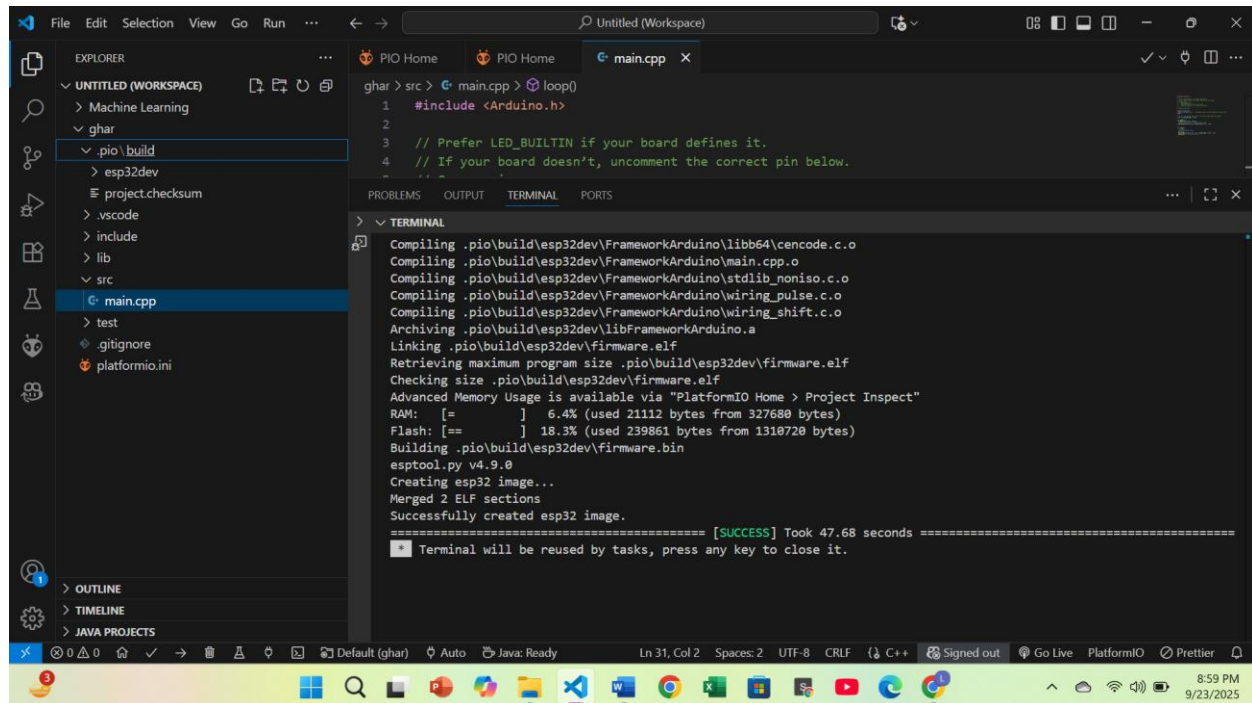
Building .pio\build\esp32dev\bootloader.bin

Generating partitions .pio\build\esp32dev\partitions.bin

Compiling .pio\build\esp32dev\FrameworkArduino\Esp.cpp.o

Name: Laiba Taseen

Reg No: 23-NTU-CS-1258



```
ghar > src > C: main.cpp > loop()
1  #include <Arduino.h>
2
3  // Prefer LED_BUILTIN if your board defines it.
4  // If your board doesn't, uncomment the correct pin below.

PROBLEMS OUTPUT TERMINAL PORTS

> TERMINAL
Compiling .pio\build\esp32dev\FrameworkArduino\libb64\cencode.c.o
Compiling .pio\build\esp32dev\FrameworkArduino\main.cpp.o
Compiling .pio\build\esp32dev\FrameworkArduino\stdlib_noniso.c.o
Compiling .pio\build\esp32dev\FrameworkArduino\wiring_pulse.c.o
Compiling .pio\build\esp32dev\FrameworkArduino\wiring_shift.c.o
Archiving .pio\build\esp32dev\libFrameworkArduino.a
Linking .pio\build\esp32dev\firmware.elf
Retrieving maximum program size .pio\build\esp32dev\firmware.elf
Checking size .pio\build\esp32dev\firmware.elf
Advanced Memory Usage is available via "PlatformIO Home > Project Inspect"
RAM: [ = ] 6.4% (used 21112 bytes from 327680 bytes)
Flash: [ == ] 18.3% (used 239861 bytes from 1310720 bytes)
Building .pio\build\esp32dev\firmware.bin
esptool.py v4.9.0
Creating esp32 image...
Merged 2 ELF sections
Successfully created esp32 image.
===== [SUCCESS] Took 47.68 seconds =====
Terminal will be reused by tasks, press any key to close it.
```